A REASONABLE ALGEBRAIC SPACE HAS A DENSE OPEN SUBSPACE WHICH IS A SCHEME

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span.

Allowed standard prerequisites (editorial).

- Locality of the schematic locus. Properties of Spaces, lemma subscheme (cross-chapter, named). Reduces to a single affine etale cover.
- Degree stratification. lemma-quasi-compact-reasonable-stratification (carded L26710). The stratum X_n and its complement.
- Finite flat equivalence global. Properties of Spaces, proposition finite-flat-equivalence-global (cross-chapter, named). The stratum piece is a scheme.
- Finite sets in affine opens. Properties, lemma ample-finite-set-in-affine (cross-chapter, named). Feeds the finite-flat-equivalence proposition.
- Nowhere dense interior. Topology, definition nowhere-dense (cross-chapter, named). Defines the interior piece X'.

Source reference. Original definition 03I8: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 1262–1262. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07S8: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 1653–1653. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01ZY: focused-seeds/geometry/S9-decent-reasonable-open-dense-scheme/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03FX: focused-seeds/geometry/S9-decent-reasonable-open-dense-scheme/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03JH: focused-seeds/geometry/S9-decent-reasonable-open-dense-scheme/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the

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Source reference. Original reference 07S6: focused-seeds/geometry/S9-decent-reasonable-open-dense-scheme/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03HN: focused-seeds/geometry/S9-decent-reasonable-open-dense-scheme/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Proposition 0.1. *Let S be a scheme. Let X be an algebraic space over S. If X is reasonable, then there exists a dense open subspace of X which is a scheme.*

Proof. By Properties of Spaces, Lemma 03JH the question is local on X. Hence we may assume there exists an affine scheme U and a surjective étale morphism $U \to X$ (Properties of Spaces, Lemma 03FX). Let n be an integer bounding the degrees of the fibres of $U \to X$ which exists as X is reasonable, see Definition 03I8. We will argue by induction on n that whenever

(1) $U \to X$ is a surjective étale morphism whose fibres have degree $\leq n$, and
(2) U is isomorphic to a locally closed subscheme of an affine scheme

then the schematic locus is dense in X.

Let $X_n \subset X$ be the open subspace which is the complement of the closed subspace $Z_{n-1} \subset X$ constructed in Lemma 07S8 using the morphism $U \to X$. Let $U_n \subset U$ be the inverse image of X_n. Then $U_n \to X_n$ is finite locally free of degree n. Hence X_n is a scheme by Properties of Spaces, Proposition 07S6 (and the fact that any finite set of points of U_n is contained in an affine open of U_n, see Properties, Lemma 01ZY).

Let $X' \subset X$ be the open subspace such that $|X'|$ is the interior of $|Z_{n-1}|$ in $|X|$ (see Topology, Definition 03HN). Let $U' \subset U$ be the inverse image. Then $U' \to X'$ is surjective étale and has degrees of fibres bounded by $n - 1$. By induction we see that the schematic locus of X' is an open dense $X'' \subset X'$. By elementary topology we see that $X'' \cup X_n \subset X$ is open and dense and we win. □